

BAR (Boletim Anual de Recursos e Reservas) vs. SEC Reserve Reporting

A Comparative Overview for the Oil & Gas Industry

Introduction

The oil and gas industry uses reserve reporting systems to estimate and communicate the quantities of hydrocarbons that can be commercially recovered. Two important frameworks are:

- **BAR (Boletim Anual de Recursos e Reservas)** – Brazil's annual reserves and resources reporting system regulated by the ANP.
- **SEC Reserve Reporting** – The reserve disclosure framework required by the U.S. Securities and Exchange Commission (SEC) for publicly traded companies.

Although both systems report oil and gas reserves, they have different objectives, audiences, and reporting requirements.

1. What is BAR?

The **Boletim Anual de Recursos e Reservas (BAR)** is an annual regulatory report submitted to Brazil's National Agency of Petroleum, Natural Gas and Biofuels (ANP).

Purpose

The BAR is used to:

- Monitor Brazil's hydrocarbon resources and reserves.
- Support regulatory oversight.
- Track field development and production activities.
- Build and maintain the official national reserves inventory.
- Support long-term energy planning.

Reporting Standard

BAR follows the **Petroleum Resources Management System (PRMS)** developed by the Society of Petroleum Engineers (SPE).

Reserve Categories

BAR commonly reports:

- **1P (Proved Reserves)**
- **2P (Proved + Probable Reserves)**
- **3P (Proved + Probable + Possible Reserves)**

In some cases, contingent and prospective resources may also be reported according to ANP requirements.

2. What is SEC Reserve Reporting?

SEC reserve reporting is a disclosure framework established by the U.S. Securities and Exchange Commission.

Purpose

SEC reporting is intended to:

- Inform investors.
- Support financial reporting.
- Provide consistency across publicly traded oil and gas companies.
- Assist analysts, lenders, and shareholders in evaluating company value.

Reporting Standard

SEC reporting follows regulations established by the SEC and focuses primarily on proved reserves.

Reserve Categories

The SEC commonly requires reporting of:

- **PDP (Proved Developed Producing)**
 - **PDNP (Proved Developed Non-Producing)**
 - **PUD (Proved Undeveloped)**
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3. Primary Differences

3.1 Objective

BAR

The primary objective is regulatory compliance and national resource management.

SEC

The primary objective is financial disclosure and investor transparency.

BAR

Regulatory reporting

Government oversight

National reserves inventory

SEC

Financial reporting

Investor disclosure

Company valuation

3.2 Classification Framework

BAR

Uses the PRMS classification system:

- 1P = Proved
- 2P = Proved + Probable
- 3P = Proved + Probable + Possible

These categories reflect increasing levels of uncertainty and potential recoverable volumes.

SEC

Focuses mainly on proved reserves:

- PDP
- PDNP
- PUD

The SEC framework generally does not emphasize probable and possible reserves in core disclosures.

3.3 Treatment of Uncertainty

BAR / PRMS

PRMS incorporates probabilistic concepts:

Category Approximate Confidence Level

1P P90

2P P50

3P P10

Where:

- **P90** = 90% probability of achieving or exceeding the estimate.
- **P50** = 50% probability.
- **P10** = 10% probability.

This approach communicates a range of potential outcomes.

SEC

SEC reporting is intentionally conservative.

Only reserves that are considered "reasonably certain" to be commercially recoverable qualify as proved reserves.

As a result:

- SEC reserve volumes are often lower than 2P or 3P volumes.
- Less emphasis is placed on upside potential.

3.4 Economic Assumptions

BAR

Reserve estimates are based on:

- Commercial viability.
- Development plans.
- Contract terms.
- PRMS commerciality principles.

SEC

The SEC requires a prescribed pricing methodology, generally based on historical average commodity prices rather than company forecast prices.

This standardization improves comparability between companies.

3.5 Development Requirements

BAR / PRMS

Projects may qualify as reserves when they satisfy PRMS technical and commercial maturity criteria.

SEC

Proved Undeveloped (PUD) reserves generally require:

- A documented development plan.
- Reasonable certainty of execution.
- Development within an expected timeframe (typically around five years, unless justified otherwise).

4. Practical Example

Assume a deepwater offshore field has the following reserve estimates:

Category Oil Reserves (MMbbl)

1P	100
2P	150
3P	220

BAR Reporting

The operator would typically report:

- 1P = 100 MMbbl
- 2P = 150 MMbbl
- 3P = 220 MMbbl

SEC Reporting

The company may report only the volumes that qualify as proved reserves under SEC definitions, such as:

- PDP
- PDNP

- PUD

The total disclosed proved reserves could be approximately 100 MMbbl, depending on development status and SEC criteria.

5. Regulatory Authorities

Brazil

Agência Nacional do Petróleo, Gás Natural e Biocombustíveis (ANP)

Responsibilities:

- Regulates exploration and production activities.
- Receives BAR submissions.
- Publishes national reserves statistics.

United States

U.S. Securities and Exchange Commission (SEC)

Responsibilities:

- Regulates public company disclosures.
 - Establishes reserve reporting requirements.
 - Protects investors through standardized reporting.
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6. Summary Comparison

Aspect	BAR (ANP)	SEC
Purpose	Regulatory oversight	Investor disclosure
Governing Authority	ANP	SEC
Reporting Standard	PRMS	SEC Regulations
Main Categories	1P, 2P, 3P	PDP, PDNP, PUD
Probabilistic Approach	Yes	Limited
Probable Reserves	Reported	Generally not emphasized

Aspect	BAR (ANP)	SEC
Possible Reserves	Reported	Generally not emphasized
Primary Audience	Government and regulators	Investors and financial markets
Geographic Focus	Brazil	SEC-registered companies worldwide

Conclusion

While both BAR and SEC reporting evaluate hydrocarbon reserves, they serve different purposes.

BAR is primarily a regulatory tool focused on resource management, national reporting, and compliance with ANP requirements using the PRMS framework. SEC reporting is designed to provide investors with standardized, conservative reserve disclosures for financial analysis and valuation.

Consequently, the same oil field may have different reported reserve volumes under BAR and SEC reporting, even though the underlying geological and engineering data are identical. The difference arises from each system's objectives, classification methodology, economic assumptions, and disclosure requirements.

Key Takeaway:

BAR = Regulatory and resource management reporting (Brazil/ANP)

SEC = Financial and investor reporting (U.S. SEC)